

Zingiber officinale rhizome as a source of 6-gingerol: isolation and pharmacological significance of the phenolic component

WIRGINIA KUKUŁA-KOCH¹, DOMINIK TARABASZ¹, APOSTOLIS ANGELIS², LEMONIA ANTONIADI², THEODOR BRAUN³, KATARZYNA GAWEL-BĘBEN⁴, THOMAS JAKSCHITZ³, ALEXIOS L. SKALTSOUNIS² and GÜNTHER K. BONN³

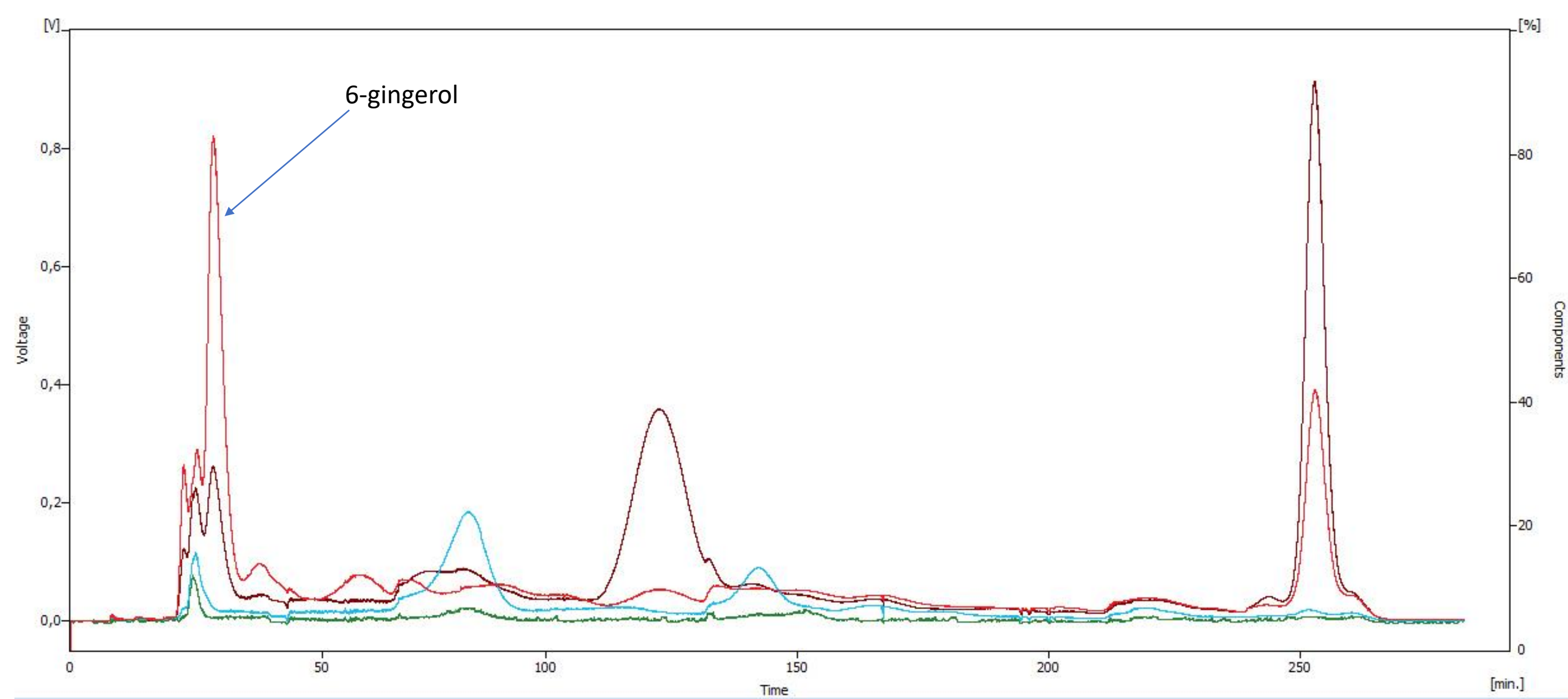
¹ Department of Pharmacognosy, Medical University of Lublin, Poland; ² Division of Pharmacognosy and Natural Products Chemistry, Department of Pharmacy, National and Kapodistrian University of Athens, Greece; ³ ADSI – Austrian Drug Screening Institute GmbH, Austria; ⁴ Department of Cosmetology, University of Information Technology and Management in Rzeszów, Poland;



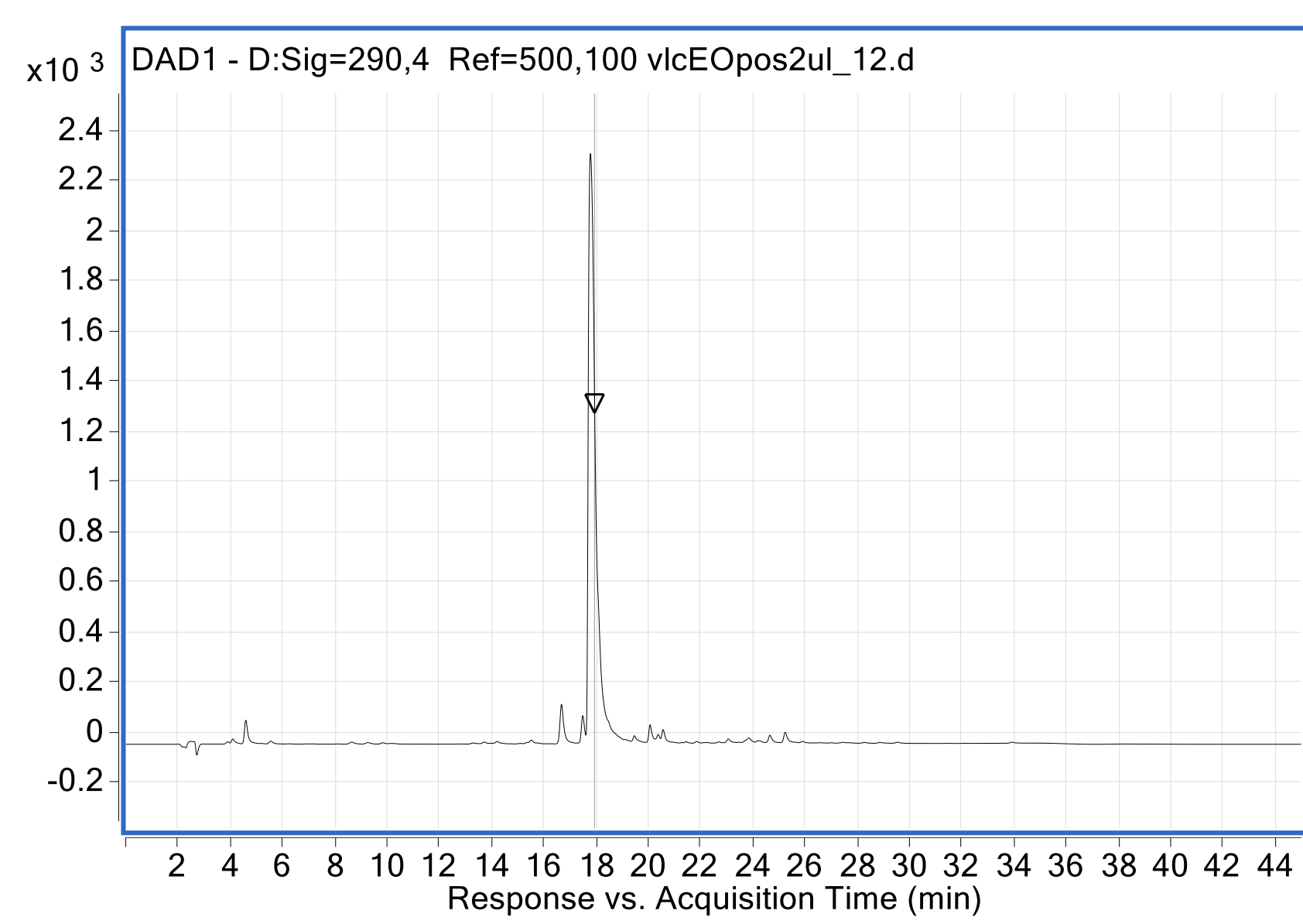
OBJECTIVE: The aim of the study was to find the beneficial isolation conditions for 6-gingerol – a pharmacologically precious phenolic constituent of the oleoresin obtained from the rhizomes of *Zingiber officinale* by means of centrifugal partition chromatography (CPC) – in a high purity form, directly from the injection of crude extract.

ISOLATION OF 6-GINGEROL BY CENTRIFUGAL PARTITION CHROMATOGRAPHY (CPC)

Centrifugal partition chromatography operated in a gradient mode was used for the recovery of 6-GIN from ginger oleoresin. The applied biphasic solvent system composed of n-hexane, ethyl acetate, methanol and water separated the phenolic constituents from terpenes abundantly represented in the sample. The recovery of 6-gingerol was achieved within 45 min of the analysis run and the obtained purity of the component exceeded 92 %.



CPC chromatogram of the methanolic extract from *Zingiber officinale*



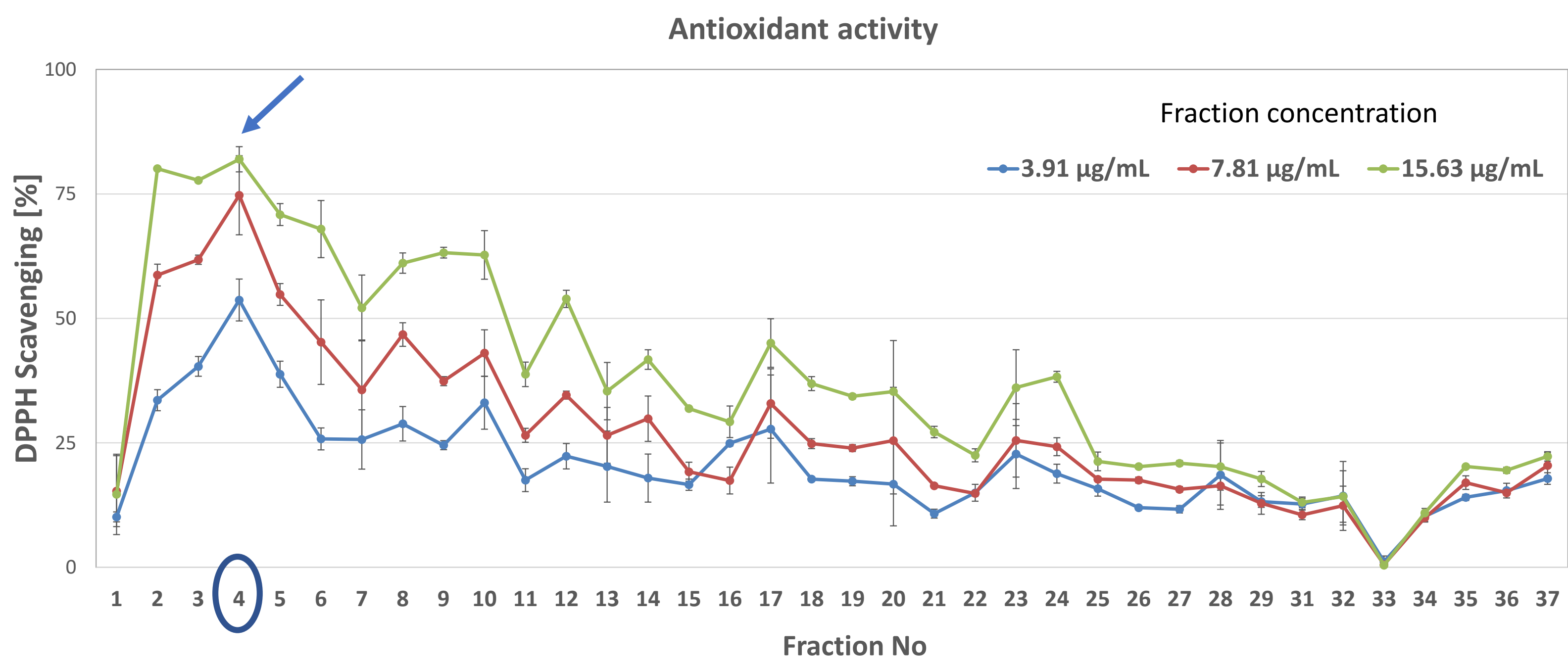
HPLC-MS chromatogram of the isolated 6-gingerol

BIOACTIVITY OF 6-GINGEROL

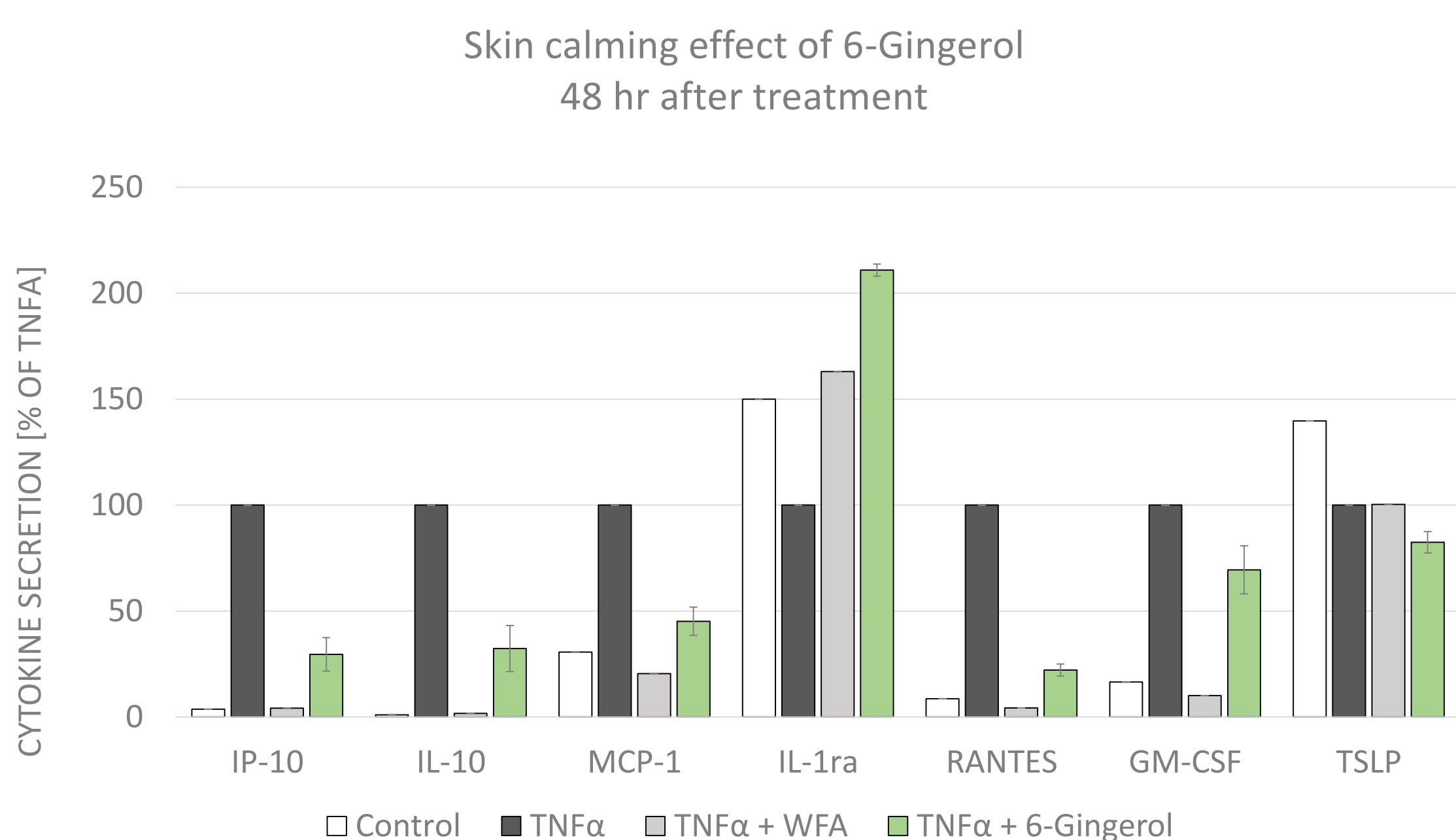
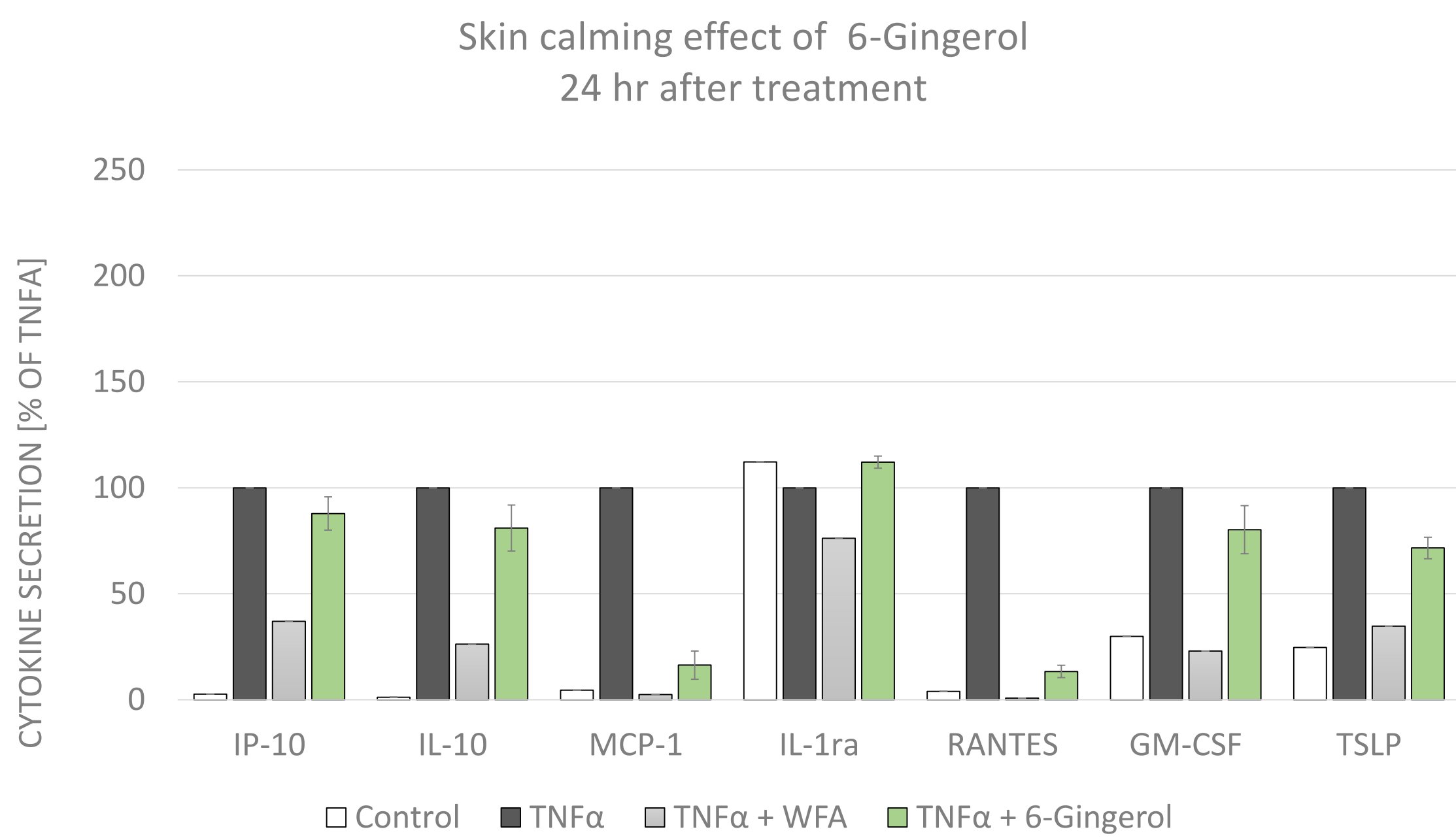
The compound was found to exhibit strong antioxidant and anti-inflammatory potential towards the following cytokines: TNF-alpha, IL-6, IP-10, IL-10, MCP-1, IL-1ra, RANTES, IL-1 alpha, GM-CSF, TSLP.

ANTIOXIDANT ACTIVITY ASSAY

The antioxidant activity determination was performed in the DPPH *in vitro* assay on all fractions obtained from the CPC chromatography. Fraction 4 (6-gingerol) was found to be the most active among the tested fractions.



ANTI-INFLAMMATORY ACTIVITY PROPERTIES of 6-GINGEROL



HaCaT keratinocytes treated for 6h with TNFα +/- WFA or 6-gingerol

RESULTS:

After 24 hours:
Significantly reduced production of RANTRS and MCP-1 (by approx. 85%), and to a small extent also TSLP (by approx. 30%)

After 48 hours:
Significantly reduced secretion of IP-10, IL-10, MCP-1, RANTES (by approx. 55-70%), slightly reduced GM-CSF (by approx. 30%) and TSLP (by approx. 18%)
Increased production of interleukin 1 receptor antagonist (IL-1ra) (by 55% compared to cells treated with TNFα)

Summary:

- ✓ CPC technique used in an optimised purification protocol provided high purity 6-gingerol directly from the injection of crude extract
- ✓ 6-gingerol was found to exhibit antioxidant and anti-inflammatory actions

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